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A multi-factor model for improved commodity pricing: calibration and an application to the oil market

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We present a new approach to commodity pricing that enhances accuracy by integrating four distinct risk factors: the spot price, stochastic volatility, convenience yield, and stochastic interest rates. We build on Yan [Valuation of commodity derivatives in a new multi-factor model. *Rev. Deriv. Res.*, 2002, 5, 251–271], the only model to our knowledge that incorporates all four sources of risk, and extend it by adding a more flexible correlation structure that captures state-dependent co-movements and time-varying risk premia. A further contribution is the explicit inclusion of the stochastic interest-rate factor within a unified Kalman-filter framework, which allows us to jointly filter the state variables and estimate model parameters using both commodity and bond market data. An empirical analysis of crude-oil futures shows that our four-factor model captures the complex dynamics of the futures term structure and consistently outperforms existing benchmarks.

Keywords: Commodity; Energy derivative; Multi-factor model; Kalman filter; Crude oil

JEL Classifications: C63, G12, Q47

1. Introduction

In recent years, the crude oil market has emerged as the world's largest and most dominant commodity market. Accurate modeling of fundamental drivers of commodity price behavior is crucial for correct pricing, hedging, and risk management of commodity derivatives.

Empirically, as documented in Litzenberger and Rabinowitz (1995), Duffie *et al.* (1999), and Eydeland and Wolyniec (2003), commodity prices are generally mean-reverting and extremely heteroscedastic. Further, Samuelson (1965) proves that the volatility of commodity futures returns diminishes with contract maturity, whereas Chiarella *et al.* (2016) and Kang *et al.* (2020) document non-monotonic, hump-shaped volatility term structures. Prices of commodity futures are often 'backwardated', meaning that they tend to decline as the delivery date approaches, although periods of contango, when futures prices exceed spot prices, are also documented (see, e.g. Almansour 2016, Nikitopoulos *et al.* 2017). Moreover, according to the analysis of crude oil prices in Routledge *et al.* (2000), the degree of

backwardation rises with volatility, and therefore volatility contains an element spanned by futures contracts.

To deal with these stylized facts, affine factor models have become popular among practitioners because they have good analytical properties and because they enable one to easily price futures and bonds. Yet, one of the major challenges is to figure out how many and what kind of factors are responsible for the spot price dynamics. In his seminal work, Schwartz (1997) considers one-, two-, and three-factor models for commodity pricing. The one-factor model attributes all uncertainty to the spot price of the commodity. However, this approach implies that returns on all futures contracts along the term structure are perfectly correlated and that the degree of backwardation remains constant over time. To address these limitations, two-factor models that incorporate both the spot price and the convenience yield have been developed, as in Gibson and Schwartz (1990) and the aforementioned work by Schwartz (1997). Thus, futures prices are no longer perfectly correlated, allowing for more complex term structures. However, Routledge *et al.* (2000) argued that the spot price-convenience yield correlation should be time-varying and two factors are not enough to capture price volatility dynamics. Consequently, various articles have proposed three-factor models, typically with either stochastic

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volatility (see Deng 2000, Geman and Nguyen 2005, Hipsproors and Jaimungal 2008, Hughen 2010, Lutz 2010, Ewald and Zou 2021) or stochastic interest rate (e.g. Cortazar and Schwartz 1994, Schwartz 1997, Cortazar and Schwartz 2003, Casassus and Collin-Dufresne 2005, Tang 2012, Mellios *et al.* 2016).

While the previous articles present evidence that effective pricing of commodity futures involves no less than three factors, filtered estimates of such models are not necessarily optimal. Hughen (2010) found that the three-factor models' pricing performance is often indistinguishable from the two-factor models' pricing performance, implying the need for a fourth factor. Cortazar and Naranjo (2006) support the empirical evidence that the introduction of a fourth factor is necessary to better fit the term structure of crude oil futures' volatility. Four-factor model specifications of Yan (2002), Schwartz and Trolle (2009a), and Schöne and Spinler (2017) seek to enhance commodity futures prices by taking into account factors such as mean reversion, convenience yield, stochastic volatility, and jump clustering, albeit in different combinations.

In this paper, to investigate and identify the appropriate number and combination of factors necessary for accurate commodity pricing, we propose a novel four-factor model that incorporates the spot price, convenience yield, interest rate, and the volatility of the spot price. Following Hughen (2010) and Schöne and Spinler (2017), we adopt affine specifications for the drift and covariance terms, and, in line with Duffee (2002), Casassus and Collin-Dufresne (2005), and Cheridito *et al.* (2007), we assume that risk premia are affine functions of the state variables. From an economic perspective, the model captures the joint dynamics between the macroeconomic environment—represented by the short-term interest rate—and the commodity-specific risk factors, namely the convenience yield and the volatility of the underlying commodity.

This formulation provides a flexible and general correlation structure among stochastic factors while retaining closed-form expressions for futures prices, which is essential for estimating the model using observable futures prices.

Several commodity-pricing models have incorporated or highlighted the role of stochastic interest rates (see, e.g. Schwartz 1997, Yan 2002, Cheng *et al.* 2018, 2019, Han *et al.* 2023). Interest-rate dynamics are central to commodity valuation because they influence the cost-of-carry, financing costs for producers, and the valuation of inventories; moreover, fluctuations in short-term rates transmit broader macroeconomic conditions into energy markets. To more effectively capture these interactions, we develop a novel joint estimation approach that explicitly incorporates bond yields within a unified Kalman filter framework. Finally, in our state-space estimation framework we use the discretization proposed by Kelly and Lord (2023) for square-root processes to ensure the state volatility process remains positive.

We perform an extensive empirical analysis, in-sample and out-of-sample, on a panel dataset of futures prices on crude oil and bond data. Our findings indicate that our approach performs better than widely used models such as Schwartz (1997), Yan (2002), Hughen (2010), and Schöne and Spinler (2017), especially when bond estimation is added,

which provides an improvement in short-term futures accuracy.

We note that a substantial stream of literature focuses on forward-price models to capture the evolution of the futures curve and to naturally accommodate unspanned stochastic volatility (see, e.g. Björk *et al.* 2006, Schwartz and Trolle 2009b, Chiarella *et al.* 2016, Cheng *et al.* 2018, Hollstein *et al.* 2020), whereas our approach instead centers on the spot price. This allows us to directly capture the fundamental dynamics of the underlying commodity, including mean-reversion and convenience yields while simplifying empirical estimation and practical implementation.

The contributions of this paper are twofold. First, we develop a four-factor model for commodity futures that jointly incorporates the stochastic dynamics of the spot price, convenience yield, stochastic volatility, and stochastic interest rates. To the best of our knowledge, only Yan (2002) considers these four sources of risk simultaneously, but his otherwise insightful framework relies on a restrictive, constant correlation structure and does not allow for state-dependent co-movements among all the factors. In contrast, our model introduces a flexible, volatility-driven correlation specification that generates time-varying interactions, both within the commodity market and between commodity and bond markets. As shown by our empirical results, this richer structure yields materially improved pricing performance relative to existing models, including the four-factor frameworks of Yan (2002) and Schöne and Spinler (2017).

Second, we introduce a joint estimation procedure that integrates futures prices and bond yields within a unified Kalman-filter framework, leveraging information from both markets. This approach yields a more comprehensive and internally consistent representation of the term structure and, as our empirical analysis shows, can in some cases improve the accuracy of commodity-futures pricing.

The remainder of this article is structured as follows: In Section 2, we present our model and derive formulas for the valuation of commodity futures and bond yields. Section 3 briefly reviews the benchmark models. Section 4 details the data sources and the model estimation framework. Empirical results, including in-sample and out-of-sample analyses as well as a hedging experiment, are presented in Section 5. Finally, Section 6 concludes the paper, while the Appendix presents the tables with the estimated parameters of the benchmark models.

2. The four-factor model

In this section, we present our proposed model for commodity prices, which incorporates Stochastic interest Rate and Volatility, yielding a total of four factors. Thus, we refer to this model as *SRV-4f*.

We assume that the commodity spot price S is driven by four independent sources of uncertainty and is influenced by the convenience yield δ , the (instantaneous) interest rate r , and the variance of the log spot price v . We define the state vector $X(t) = (\ln S(t) \ \delta(t) \ r(t) \ v(t))^T$, which we assume satisfies

the following stochastic differential equation, under the risk-neutral probability measure:

$$dX(t) = [A + BX(t)] dt + \Sigma^{1/2}(t, X(t)) dZ(t),$$

$$A = \begin{pmatrix} 0 \\ k_2\mu_2 \\ k_3\mu_3 \\ k_4\mu_4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 & +1 & -\frac{1}{2} \\ 0 & -k_2 & 0 & 0 \\ 0 & 0 & -k_3 & 0 \\ 0 & 0 & 0 & -k_4 \end{pmatrix}, \quad (1)$$

where $dZ(t)$ is a vector of four uncorrelated standard Brownian motion increments, $\Sigma^{1/2}(t, X(t))$ denotes the Cholesky decomposition of the instantaneous variance and covariance matrix given by

$$\Sigma(t, X(t)) = \Omega_0 + \Omega_1 v(t), \quad (2)$$

where

$$\Omega_0 = \begin{pmatrix} 0 & s_{12} & s_{13} & 0 \\ s_{12} & s_{22} & s_{23} & 0 \\ s_{13} & s_{23} & s_{33} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\Omega_1 = \begin{pmatrix} 1 & \rho_{12}\sigma_{22} & \rho_{13}\sigma_{33} & \rho_{14}\sigma_{44} \\ \rho_{12}\sigma_{22} & \sigma_{22}^2 & \rho_{23}\sigma_{22}\sigma_{33} & \rho_{24}\sigma_{22}\sigma_{44} \\ \rho_{13}\sigma_{33} & \rho_{23}\sigma_{22}\sigma_{33} & \sigma_{33}^2 & \rho_{34}\sigma_{33}\sigma_{44} \\ \rho_{14}\sigma_{44} & \rho_{24}\sigma_{22}\sigma_{44} & \rho_{34}\sigma_{33}\sigma_{44} & \sigma_{44}^2 \end{pmatrix}.$$

As in Duffie and Kan (1996), both the drift term and $\Sigma(t, X(t))$ are affine functions of the state variables. We note that $\Omega_0(1, 1) = 0$ and $\Omega_1(1, 1) = 1$, so that $v(t)$ is actually the variance of the first state variable $\log S(t)$, and we require $\Omega_0 + \Omega_1 v(t)$ to be positive definite.

The stochastic volatility factor v_t enters the diffusion terms of all state variables. This specification is supported by both economic and empirical arguments. Volatility shocks in commodity markets reflect changes in macroeconomic uncertainty, inventory risk, and information flow, all of which simultaneously affect spot prices, convenience yields, and interest rates. Empirically, these variables exhibit pronounced co-movements in their conditional volatilities, as documented by Routledge *et al.* (2000) and Casassus and Collin-Dufresne (2005). In addition, our formulation naturally accommodates potential interdependence between the volatility of the interest rate $r(t)$ and that of the spot price $S(t)$, in a manner analogous to the discrete-time framework proposed by Ballestra *et al.* (2024). Consequently, v_t serves as a market-wide volatility factor that governs the joint heteroskedasticity of both commodity and financial risks.

A well-documented feature of commodity markets is that risk premia may vary over time in response to shifting inventory conditions, macroeconomic cycles, and volatility regimes. When inventories are low, marginal convenience yields increase and producers face heightened supply risk, leading to larger, state-dependent premia for bearing spot-price and convenience-yield risk (see Gibson and Schwartz 1990). Moreover, macroeconomic conditions and changes in financial market volatility transmit into commodity risk premia through hedging pressure

and time-varying investment flows, implying that premia must depend on the underlying state variables to capture these dynamic cross-market linkages (see Kang *et al.* 2020). Then, we define the vector of risk-premia $\Lambda(t, X(t)) = (\lambda_x(t, S(t)) \ \lambda_\delta(t, \delta(t)) \ \lambda_r(t, r(t)) \ \lambda_v(t, v(t)))^\top$ as an affine function of $X(t)$, that is

$$\Lambda(t, X(t)) = \Sigma(t, X(t))^{-1} (\hat{A} - A + [\hat{B} - B]X(t)), \quad (3)$$

where the physical drift parameters $\hat{A}$ and $\hat{B}$ are given by

$$\hat{A} = \begin{pmatrix} \hat{\mu}_1 \\ \hat{k}_2\hat{\mu}_2 \\ \hat{k}_3\hat{\mu}_3 \\ \hat{k}_4\hat{\mu}_4 \end{pmatrix}, \quad \hat{B} = \begin{pmatrix} 0 & -1 & 0 & -\frac{1}{2} \\ 0 & -\hat{k}_2 & 0 & 0 \\ 0 & 0 & -\hat{k}_3 & 0 \\ 0 & 0 & 0 & -\hat{k}_4 \end{pmatrix}. \quad (4)$$

From the Girsanov Theorem, see Theorem 11.3 in Björk (2009), we know that the process $Z(t)^\mathbb{P}$, defined by

$$Z(t)^\mathbb{P} = Z(t) - \int_0^t \Lambda(s, X(s)) ds, \quad (5)$$

is a standard $\mathbb{P}$ -Wiener process, where $\mathbb{P}$ denotes the physical probability measure.

Substituting (5) into (1) yields the continuous time model dynamics under $\mathbb{P}$:

$$dX(t) = [\hat{A} + \hat{B}X(t)] dt + \Sigma^{1/2}(t, X(t)) dZ^\mathbb{P}(t). \quad (6)$$

From a mathematical standpoint, the diffusion term in (2) follows an affine specification similar to the elegant model proposed by Schöne and Spinler (2017). However, we model different risk factors and we allow for a more parsimonious parametrization of the expressions for B and Ω_0 (see Section 3.6).

2.1. Futures prices

The futures price at time t with time to maturity $\tau = T - t$ can be computed as the risk-neutral expectation of the futures price:

$$F(t, \tau, X(t)) = \mathbb{E} [e^{\ln S(t+\tau)} | \mathcal{F}(t)],$$

where $\mathcal{F}(t)$ denotes the information available at time t . The futures price is then the solution to the following partial differential equation (PDE) (see Proposition 5.5 in Björk 2009):

$$\frac{\partial F(t, \tau, X(t))}{\partial \tau} = (\nabla_x F(t, \tau, X(t)))^\top [A + BX(t)] + \frac{1}{2} \text{Tr} \{ (H_x F(t, \tau, X(t))) \Sigma(t, X(t)) \}, \quad (7)$$

with the terminal condition $F(t, 0, X(t)) = e^{X(t)}$, where $\nabla_x F(t, \tau, X(t))$ and $H_x F(t, \tau, X(t))$ are the gradient of $F(t, \tau, X(t))$ with respect to $X(t)$ and the Hessian matrix of $F(t, \tau, X(t))$ with respect to $X(t)$, respectively, and Tr is the trace operator.

Given that the drift and diffusion terms in (1) are affine functions of $X(t)$, the solution to equation (7) can be guessed as follows:

$$\ln F(t, \tau, X(t)) = \alpha(\tau) + \beta(\tau)X(t), \quad (8)$$

where $\beta(\tau) = (\beta_1(\tau) \ \beta_2(\tau) \ \beta_3(\tau) \ \beta_4(\tau))$. By substituting equation (8) into (7) we obtain

$$\begin{aligned} \frac{d\alpha(\tau)}{d\tau} + \frac{d\beta(\tau)}{d\tau}X(t) &= \beta(\tau)[A + BX(t)] \\ &+ \frac{1}{2} [\beta(\tau)\Omega_0\beta(\tau)^\top + \beta(\tau)\Omega_1\beta(\tau)^\top v(t)]. \end{aligned}$$

Separation of variables leads to the following system of ordinary differential equations (ODEs):

$$\begin{aligned} \frac{d\alpha(\tau)}{d\tau} &= \beta(\tau)A + \frac{1}{2}\beta(\tau)\Omega_0\beta(\tau)^\top, \\ \frac{d\beta_1(\tau)}{d\tau} &= \beta(\tau)B_{(1)}, \quad \frac{d\beta_2(\tau)}{d\tau} = \beta(\tau)B_{(2)}, \\ \frac{d\beta_3(\tau)}{d\tau} &= \beta(\tau)B_{(3)}, \\ \frac{d\beta_4(\tau)}{d\tau} &= \beta(\tau)B_{(4)} + \frac{1}{2}\beta(\tau)\Omega_1\beta(\tau)^\top, \end{aligned} \quad (9)$$

where $B_{(i)}$ denotes the i th column of B , $i = 1, 2, 3, 4$. Equations (9) must be solved with terminal conditions $\alpha(0) = 0$, $\beta_1(0) = 1$, $\beta_2(0) = 0$, $\beta_3(0) = 0$ and $\beta_4(0) = 0$.

2.2. Bond yields

The present value at time t of a unit discount bond $P(t, \tau, X(t))$ with time to maturity τ can be computed as:

$$P(t, \tau, X(t)) = \mathbb{E} \left[e^{-\int_t^{\tau} r(s) ds} \mid \mathcal{F}(t) \right]. \quad (10)$$

The conditional expectation (10) is the solution of the following PDE:

$$\begin{aligned} P(t, \tau, X(t))r(t) &= -\frac{\partial P(t, \tau, X)}{\partial \tau} \\ &+ (\nabla_x P(t, \tau, X(t)))^\top [A + BX(t)] \\ &+ \frac{1}{2} \text{Tr} \{ (H_x P(t, \tau, X(t))) \Sigma(t, X(t)) \}, \end{aligned} \quad (11)$$

with terminal condition $P(t, 0, X(t)) = 1$. The PDE (11) can be solved as

$$\ln P(t, \tau, X(t)) = \gamma(\tau) + \zeta(\tau)X(t), \quad (12)$$

where $\zeta(\tau) = (\zeta_1(\tau) \ \zeta_2(\tau) \ \zeta_3(\tau) \ \zeta_4(\tau))$. By substituting equation (12) into (11) we obtain

$$\frac{d\gamma(\tau)}{d\tau} + \frac{d\zeta(\tau)}{d\tau}X(t)$$

$$\begin{aligned} &= \zeta(\tau)[A + BX(t)] \\ &+ \frac{1}{2} [\zeta(\tau)\Omega_0\zeta(\tau)^\top + \zeta(\tau)\Omega_1\zeta(\tau)^\top v(t)] - r(t), \end{aligned}$$

which leads to the following system of ODEs:

$$\begin{aligned} \frac{d\gamma(\tau)}{d\tau} &= \zeta(\tau)A + \frac{1}{2}\zeta(\tau)\Omega_0\zeta(\tau)^\top, \\ \frac{d\zeta_1(\tau)}{d\tau} &= \zeta(\tau)B_{(1)}, \quad \frac{d\zeta_2(\tau)}{d\tau} = \zeta(\tau)B_{(2)}, \\ \frac{d\zeta_3(\tau)}{d\tau} &= \zeta(\tau)B_{(3)} - 1, \\ \frac{d\zeta_4(\tau)}{d\tau} &= \zeta(\tau)B_{(4)} + \frac{1}{2}\zeta(\tau)\Omega_1\zeta(\tau)^\top, \end{aligned}$$

and $\gamma(0) = \zeta(0) = 0$ so that $P(t, 0, X(t)) = 1$.

3. Benchmark models

In this section, for the readers' convenience, we briefly recall well-known factor models for commodity prices which we will utilize as benchmarks.

3.1. Schwartz's (1997) one-factor model

The one-factor model of Schwartz (1997), hereafter *SCH-1f*, assumes that $\ln S(t)$ follows an Ornstein-Uhlenbeck stochastic process, under the risk-neutral measure:

$$d \ln S(t) = \kappa (\alpha - \lambda - \ln S(t)) dt + \sigma dZ(t),$$

where λ is the (constant) market price of risk and $dZ(t)$ denotes the increment to a standard Brownian motion.

According to equation (8) of Schwartz (1997), we can compute the logarithm of the futures price with time to maturity τ as:

$$\begin{aligned} \ln F(t, \tau, S(t)) &= e^{-\kappa\tau} \ln S(t) + (1 - e^{-\kappa\tau})(\alpha - \lambda) \\ &+ \frac{\sigma^2}{4\kappa} (1 - e^{-2\kappa\tau}). \end{aligned}$$

3.2. Schwartz's (1997) two-factor model

The two-factor model of Schwartz (1997), denoted *SCH-2f*, models the logarithm of the spot price and convenience yield δ via the following equations, under the risk-neutral measure:

$$\begin{aligned} d \ln S(t) &= (r - \delta(t) - \frac{1}{2}\sigma^2) dt + \sigma_1 dZ_1(t), \\ d\delta(t) &= \kappa[(\alpha - \delta(t)) - \lambda]dt + \sigma_2 dZ_2(t), \end{aligned}$$

where the increments of the standard Brownian motions are correlated with $\mathbb{E}[dZ_1(t) dZ_2(t)] = \rho dt$. The logarithm of the futures price with time to maturity τ is given as in equation (19) of Schwartz (1997):

$$\ln F(t, \tau, S(t), \delta(t)) = \ln S(t) - \delta(t) \frac{1 - e^{-\kappa\tau}}{\kappa} + A^*(\tau),$$

where $A^*(\tau)$ is specified in equation (20) of Schwartz (1997).

3.3. Schwartz's (1997) stochastic interest rate model

The three-factor model of Schwartz (1997), which we denote by *SCH-3f*, comprises the following stochastic differential equations under the risk-neutral measure:

$$\begin{aligned} d \ln S(t) &= (r(t) - \delta(t) - \frac{1}{2}\sigma^2) dt + \sigma_1 dZ_1(t), \\ d\delta(t) &= \kappa (\hat{\alpha} - \delta(t)) dt + \sigma_2 dZ_2(t), \\ dr(t) &= a (\hat{\mu} - r(t)) dt + \sigma_3 dZ_3(t), \end{aligned}$$

where the Brownian motion increments are correlated via $\mathbb{E}[dZ_1(t) dZ_2(t)] = \rho_1 dt$, $\mathbb{E}[dZ_2(t) dZ_3(t)] = \rho_2 dt$ and $\mathbb{E}[dZ_1(t) dZ_3(t)] = \rho_3 dt$. The logarithm of the futures price with time to maturity τ is given by equation (27) of Schwartz (1997):

$$\begin{aligned} \ln F(t, \tau, S(t), \delta(t), r(t)) &= \ln S(t) - \frac{\delta(t)(1 - e^{-\kappa\tau})}{\kappa} \\ &\quad + \frac{r(t)(1 - e^{-a\tau})}{a} + C^*(\tau), \end{aligned}$$

where $C^*(\tau)$ is specified in equation (28) of Schwartz (1997).

3.4. Hughen's (2010) stochastic volatility model

The model of Hughen (2010), hereafter *HU-3f*, considers the state vector $X(t) = (\ln S(t) \ \delta(t) \ v(t))^\top$, which, under the risk-neutral measure, satisfies the following SDE:

$$\begin{aligned} dX(t) &= (A + BX(t)) dt + \Sigma^{1/2}(t, X(t)) dZ(t), \\ \Sigma(t, X(t)) &= \Omega_0 + \Omega_1 v(t), \end{aligned}$$

where $Z(t)$ is a vector of three uncorrelated Brownian motions, and

$$\begin{aligned} A &= \begin{pmatrix} 0 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & -1/2 \\ \kappa_{21} & \kappa_{22} & \kappa_{23} \\ 0 & 0 & \kappa_{33} \end{pmatrix}, \\ \Omega_0 &= \begin{pmatrix} 0 & s_{12} & \sigma_{13}\vartheta \\ s_{12} & s_{22} & \sigma_{23}\vartheta \\ \sigma_{13}\vartheta & \sigma_{23}\vartheta & \sigma_{33}\vartheta \end{pmatrix}, \quad \Omega_1 = \begin{pmatrix} 1 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{pmatrix}, \end{aligned}$$

where Ω_1 is positive definite and $\Omega_0 - \Omega_1\vartheta$ is positive semi-definite with $\vartheta \leq 0$. The logarithm of the futures price with time to maturity τ is given by equation (27) of Hughen (2010), that is:

$$\ln F(t, \tau, X(t)) = \alpha(\tau) + \beta(\tau)X(t),$$

where $\alpha(\tau)$ and $\beta(\tau) = (\beta_1(\tau) \ \beta_2(\tau) \ \beta_3(\tau))$ can be obtained via numerical integration of the following system of ODEs:

$$\begin{aligned} \frac{d\alpha(\tau)}{d\tau} &= \beta(\tau)A + \frac{1}{2}\beta(\tau)\Omega_0\beta(\tau)^\top, \quad \frac{d\beta_1(\tau)}{d\tau} = \beta(\tau)B_{(1)}, \\ \frac{d\beta_2(\tau)}{d\tau} &= \beta(\tau)B_{(2)}, \\ \frac{d\beta_3(\tau)}{d\tau} &= \beta(\tau)B_{(3)} + \frac{1}{2}\beta(\tau)\Omega_1\beta(\tau)^\top, \end{aligned}$$

where $B_{(i)}$ denotes the i th column of B , $i = 1, 2, 3$, and the terminal conditions are $\alpha(0) = 0$, $\beta_1(0) = 1$, $\beta_2(0) = 0$, $\beta_3(0) = 0$.

3.5. Yan's (2002) stochastic volatility with jumps model

The four-factor model of Yan (2002), which we denote by *YAN-4f*, comprises the following set of equations under the risk-neutral measure:

$$\begin{aligned} d \ln S(t) &= \left(r(t) - \delta(t) - v\mu_J - \frac{\sigma_x^2}{2} - \frac{1}{2}v(t) \right) dt \\ &\quad + \sigma_x dZ_x(t) + \sqrt{v(t)} dZ_v(t) + J dq, \\ d\delta(t) &= (\mu_\delta - \kappa_\delta \delta(t)) dt + \sigma_\delta dZ_\delta(t), \\ dr(t) &= (\mu_r - \kappa_r r(t)) dt + \sigma_r \sqrt{r(t)} dZ_r(t), \\ dv(t) &= (\mu_v - \kappa_v v(t)) dt + \sigma_v \sqrt{v(t)} dZ_v(t) + J_v dq, \end{aligned}$$

where q is a Poisson process with constant intensity v , J and $J_v \sim \text{Exp}(\theta)$, $\theta > 0$ denotes the jump size of the log spot price and the volatility, respectively, with $\ln(1 + J) \sim N\left(\ln(1 + \mu_J) - \frac{\sigma_J^2}{2}, \sigma_J^2\right)$. The Brownian motion increments are correlated as $\mathbb{E}[dZ_x(t) dZ_\delta(t)] = \rho_{x\delta} dt$ and $\mathbb{E}[dZ_x(t) dZ_v(t)] = \rho_{xv} dt$. The logarithm of the futures price with time to maturity τ is given by:

$$\begin{aligned} \ln F(t, \tau, S(t), \delta(t), r(t)) &= \ln S(t) + \beta_0(\tau) \\ &\quad + \beta_\delta(\tau)\delta(t) + \beta_r(\tau)r(t), \end{aligned}$$

where the expressions for $\beta_0(\tau)$, $\beta_\delta(\tau)$ and $\beta_r(\tau)$ are given in equation (13) of Yan (2002). We note that the futures price does not directly depend on the spot volatility. Moreover, only the convenience yield and the spot volatility are correlated with the return process, whereas all the other correlations are set to zero.

3.6. Schöne and Spinler (2017) four-factor model

The four-factor model of Schöne and Spinler (2017), denoted *SS-4f*, uses as state vector $X(t) = (\ln S(t) \ \theta(t) \ q(t) \ v(t))^\top$, where $\theta(t)$ represents the long-term level of $\ln S(t)$ and $q(t)$ is a factor related to the cost of carry. Schöne and Spinler (2017) formulate their model, under the risk-neutral measure, as follows:

$$\begin{aligned} dX(t) &= [A + BX(t)] dt + \Sigma^{1/2}(t, X(t)) dZ(t), \\ \Sigma(t, X(t)) &= \Omega_0 + \Omega_1 v(t), \end{aligned}$$

where $Z(t)$ is a vector of four uncorrelated Brownian motions and

$$\begin{aligned} A &= \begin{pmatrix} 0 \\ \mu_2 \\ \kappa_{33}\mu_3 \\ \kappa_{44}\mu_4 \end{pmatrix}, \quad B = \begin{pmatrix} -\kappa_{11} & \kappa_{11} & -1 & \kappa_{14} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\kappa_{33} & 0 \\ 0 & 0 & 0 & -\kappa_{44} \end{pmatrix}, \\ \Omega_0 &= \begin{pmatrix} 0 & s_{12} & s_{13} & \vartheta\sigma_{14} \\ s_{12} & s_{22} & s_{23} & \vartheta\sigma_{24} \\ s_{13} & s_{23} & s_{33} & \vartheta\sigma_{34} \\ \vartheta\sigma_{14} & \vartheta\sigma_{24} & \vartheta\sigma_{34} & \vartheta\sigma_{44} \end{pmatrix}, \\ \Omega_1 &= \begin{pmatrix} 1 & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} & \sigma_{24} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} & \sigma_{34} \\ \sigma_{14} & \sigma_{24} & \sigma_{34} & \sigma_{44} \end{pmatrix}, \end{aligned}$$

where Ω_1 and $\Omega_0 - \Omega_1 \vartheta$ are positive semi-definite with $\vartheta \leq 0$. The logarithm of the futures price with time to maturity τ is given by equation (11) of Schöne and Spinler (2017), that is

$$\ln F(t, \tau, X(t)) = \alpha(\tau) + \beta(\tau)X(t),$$

where $\alpha(\tau)$ and $\beta(\tau) = (\beta_1(\tau) \ \beta_2(\tau) \ \beta_3(\tau) \ \beta_4(\tau))$ can be obtained via numerical integration of the following system of differential equations:

$$\begin{aligned} \frac{d\alpha(\tau)}{d\tau} &= \beta(\tau)A + \frac{1}{2}\beta(\tau)\Omega_0\beta(\tau)^\top, \\ \frac{d\beta_1(\tau)}{d\tau} &= \beta(\tau)B_{(1)}, \quad \frac{d\beta_2(\tau)}{d\tau} = \beta(\tau)B_{(2)}, \\ \frac{d\beta_3(\tau)}{d\tau} &= \beta(\tau)B_{(3)}, \\ \frac{d\beta_4(\tau)}{d\tau} &= \beta(\tau)B_{(4)} + \frac{1}{2}\beta(\tau)\Omega_1\beta(\tau)^\top, \end{aligned}$$

where $B_{(i)}$ denotes the i th column of B , $i = 1, 2, 3, 4$, and the terminal solutions are $\alpha(0) = \beta_2(0) = \beta_3(0) = \beta_4(0) = 0$ and $\beta_1(0) = 1$.

The *SS-4f* employs four factors as our *SRV-4f* model. However, while *SS-4f* models the log spot price together with latent ‘long-term level’ and ‘cost-of-carry’ factors, our model replaces these with economically interpretable convenience-yield and stochastic-interest-rate factors. Moreover, our affine structure employs a more parsimonious specification for matrices B and Ω_0 .

4. Estimation methodology

Since the variables included in the state vector are typically unobserved, we employ the estimation framework of Duan and Simonato (1999), which is based on the Kalman filter. This approach enables the extraction of latent variables from a cross-section of observables, while simultaneously estimating the parameters of the commodity price models (see Schwartz 1997, Schöne and Spinler 2017). Within this Kalman filter estimation procedure, the empirical spot price $S(t)$ is not used.

In section 4.1, we illustrate the traditional Kalman filter equations for estimation procedures that rely solely on observed futures prices. In section 4.2, we derive a state-space system and the related filtering equations, which allow for the joint estimation of futures and bond prices. Lastly, section 4.3 derives a discretized transition equation for the volatility process, ensuring its positivity.

4.1. Futures prices

At each time t we observe the prices of H futures with different maturities τ_i , for $i = 1, \dots, H$. To proceed, we stack the corresponding pricing equations, derived from (8) by adding a measurement error term, resulting in the following

representation:

$$\begin{aligned} y(t) &= \begin{pmatrix} \ln F^{(1)}(t, \tau_1, X(t)) \\ \vdots \\ \ln F^{(H)}(t, \tau_H, X(t)) \end{pmatrix} \\ &= \alpha(\tau) + \beta(\tau)X(t) + \begin{pmatrix} \varepsilon^{(1)}(t) \\ \vdots \\ \varepsilon^{(H)}(t) \end{pmatrix}, \end{aligned} \quad (13)$$

where

$$\begin{aligned} \alpha(\tau) &= (\alpha(\tau_1) \ \dots \ \alpha(\tau_H))^\top, \quad \beta(\tau) \\ &= \begin{pmatrix} \beta_1(\tau_1) & \beta_2(\tau_1) & \beta_3(\tau_1) & \beta_4(\tau_1) \\ \vdots & \vdots & \vdots & \vdots \\ \beta_1(\tau_H) & \beta_2(\tau_H) & \beta_3(\tau_H) & \beta_4(\tau_H) \end{pmatrix} \end{aligned}$$

and $\varepsilon^{(i)}(t)$ represents the measurement error related to the logarithm of futures price i with zero mean and standard deviation σ_{ε_i} , $i = 1, \dots, H$. This accounts for the possibility of bid-ask spreads, nonsimultaneity of observations and errors in the data. In a correctly specified model, the errors $\varepsilon(t)$ should be serially and cross-sectionally uncorrelated with zero mean.

To obtain the transition equation for the state-space system we need to derive the expressions for the conditional mean and variance of the (unobserved) state variables over a discrete time interval of length h , which we set equal to $1/252$ to represent daily data. We discretize equation (6) over h using Euler discretization and we obtain the following transition equation:

$$\tilde{X}(t+h) = \hat{A}h + (I_4 + \hat{B}h)\tilde{X}(t) + \sqrt{h}\Sigma^{1/2}(t, \tilde{X}(t))\eta(t+h), \quad (14)$$

where $\tilde{X}(t)$ represents the discretized approximation of the process at time step t , $\eta(t+h)$ is a normally distributed (4×1) error vector with zero mean and identity covariance, and I_4 denotes the (4×4) identity matrix. Given that $\Sigma(t, X(t))$ is a function of $X(t)$, the transition density of equation (14) will not be Gaussian, so that we have a quasi-optimal Kalman filter. However, as explained in Duan and Simonato (1999) and in Ewald and Zou (2021), the use of this quasi-optimal filter yields an approximate quasi-likelihood function with which parameter estimation can be efficiently carried out.

The state-space system of equations (13)–(14) can be directly used in the Kalman filter recursion, which we briefly recall hereafter. Using the set of parameters $\theta_F = \{A, B, \hat{A}, \hat{B}, \Omega_0, \Omega_1, \sigma_{\varepsilon_1}, \dots, \sigma_{\varepsilon_H}\}$, we compute the one-step ahead prediction and variance of $X(t)$ for a step size h , conditional on the information at time t :

$$\begin{aligned} \tilde{X}(t+h|t) &= \hat{A}h + [I_4 + \hat{B}h]\tilde{X}(t|t), \\ P(t+h|t) &= [I_4 + \hat{B}h]P(t|t)[I_4 + \hat{B}h]^\top + \Sigma(t, \tilde{X}(t)|t), \end{aligned} \quad (15)$$

where

$$\Sigma(t, \tilde{X}(t)|t) = \Omega_0 + \Omega_1\tilde{v}(t|t).$$

To finish the Kalman filter we need the updating equations:

$$\begin{aligned}\tilde{X}(t|t) &= \tilde{X}(t|t-h) + P(t|t-h)\beta(\tau)^\top V(t|t-h)^{-1}e(t), \\ P(t|t) &= P(t|t-h) - P(t|t-h)\beta(\tau)^\top, \\ V(t|t-h)^{-1}\beta(\tau)P(t|t-h),\end{aligned}$$

where $e(t) = y(t) - y(t|t-h)$ is the forecast error. We compute then the one-period-ahead prediction and variance of the logarithm of futures prices as:

$$y(t+h|t) = \alpha(\tau) + \beta(\tau)\tilde{X}(t+h|t), \quad (16)$$

$$V(t+h|t) = \beta(\tau)P(t+h|t)\beta(\tau)^\top + Q, \quad (17)$$

where Q is a $(H \times H)$ diagonal matrix with entries $\sigma_{\varepsilon_i}^2$, $i = 1, \dots, H$. The estimated parameter vector θ_F^* solves

$$\theta_F^* = \arg \max_{\theta} \sum_{t=1}^N \log L(\theta | e(t), V(t|t-h)),$$

where N represents the length of the time series of futures prices and the day- t log-likelihood is equal to

$$\begin{aligned}\log L(\theta | e(t), V(t|t-h)) \\ = -\frac{1}{2} (N \log 2\pi + [\log \det (V(t|t-h)) \\ + e(t)^\top V^{-1}(t|t-h)e(t)]). \quad (18)\end{aligned}$$

4.2. Futures prices and bond yields

So far, we have shown how to estimate the model using only futures prices. As explained in Schwartz (1997), the parameters θ_F of the state vector should be estimated simultaneously from a time-series cross-sectional dataset of futures prices and bond yields. In this section, we illustrate an estimation framework based on a state-space representation that jointly models futures prices and bond yields, allowing for effective estimation of the model parameters.

Let $R(t, \tau, X(t))$ denote the continuously compounded yield at time t on a zero-coupon bond of maturity τ with price $P(t, \tau, X(t))$:

$$R(t, \tau, X(t)) = -\frac{1}{\tau} \ln P(t, \tau, X(t)). \quad (19)$$

Similarly to our approach for futures prices, we assume that yields for different maturities are observed with errors of unknown magnitudes. Using the bond pricing formula in equation (12), the yield to maturity can be written, after the addition of a measurement error to equation (19), as:

$$R(t, \tau, X(t)) = -\frac{1}{\tau} [\gamma(\tau) + \zeta(\tau)X(t)] + \psi(t), \quad (20)$$

where $\psi(t)$ is an error term with zero mean and standard deviation σ_ψ .

Given that at each time t we observe H futures and K bond yields with different maturities, equations (13) and (20) can

be stacked to obtain the following representation:

$$\begin{aligned}y(t) &= \begin{pmatrix} \ln F^{(1)}(t, \tau_1, X(t)) \\ \vdots \\ \ln F^{(H)}(t, \tau_H, X(t)) \\ R^{(1)}(t, \tau_1, X(t)) \\ \vdots \\ R^{(K)}(t, \tau_K, X(t)) \end{pmatrix} \\ &= \begin{pmatrix} \alpha(\tau) \\ \gamma(\tau) \end{pmatrix} + \begin{pmatrix} \beta(\tau) \\ \zeta(\tau) \end{pmatrix} X(t) + \begin{pmatrix} \varepsilon^{(1)}(t) \\ \vdots \\ \varepsilon^{(H)}(t) \\ \psi^{(1)}(t) \\ \vdots \\ \psi^{(K)}(t) \end{pmatrix}. \quad (21)\end{aligned}$$

where

$$\begin{aligned}\gamma(\tau) &= \left(-\frac{\gamma(\tau_1)}{\tau_1} \quad \dots \quad -\frac{\gamma(\tau_K)}{\tau_K} \right)^\top, \quad \zeta(\tau) \\ &= \begin{pmatrix} -\frac{\zeta_1(\tau_1)}{\tau_1} & -\frac{\zeta_2(\tau_1)}{\tau_1} & -\frac{\zeta_3(\tau_1)}{\tau_1} & -\frac{\zeta_4(\tau_1)}{\tau_1} \\ \vdots & \vdots & \vdots & \vdots \\ -\frac{\zeta_1(\tau_K)}{\tau_K} & -\frac{\zeta_2(\tau_K)}{\tau_K} & -\frac{\zeta_3(\tau_K)}{\tau_K} & -\frac{\zeta_4(\tau_K)}{\tau_K} \end{pmatrix},\end{aligned}$$

and $\psi^{(j)}(t)$ represents the measurement error related to bond yield j at time t with zero mean and standard deviation σ_{ψ_j} , $j = 1, \dots, K$. Moreover, given that now we also observe bond yields, we need to consider the parameter set $\theta_{FB} = \{A, B, \hat{A}, \hat{B}, \Omega_0, \Omega_1, \sigma_{\varepsilon_1}, \dots, \sigma_{\varepsilon_H}, \sigma_{\psi_1}, \dots, \sigma_{\psi_K}\}$. Note that θ_{FB} contains the same parameters as θ_F along with the standard deviations σ_{ψ_j} , $j = 1, \dots, K$. In addition, we adjust Q of the observation equation (17) as a $(H+K) \times (H+K)$ diagonal matrix with entries $\sigma_{\varepsilon_i}^2$ and $\sigma_{\psi_j}^2$, $i = 1, \dots, H$, $j = 1, \dots, K$.

4.3. Positivity of the state volatility process

The Euler discretization scheme in equation (14) does not guarantee the positivity of the CIR process for the variance process. We address this issue by modifying equation (14) to utilize an alternative discretization approach. This adjustment ensures the positivity of the variance state process without enforcing restrictive parameter conditions. We follow the numerical approach developed in Kelly and Lord (2023), which is to apply a Lamperti transformation to the CIR process and to numerically approximate the related process.

Let us consider the equation for the variance process v_t in (6):

$$dv(t) = \widehat{\kappa}_4(\widehat{\mu}_4 - v(t)) dt + \sqrt{v(t)} \Omega_1^{1/2} dZ(t), \quad (22)$$

where $\Omega_1^{1/2}$ denotes the fourth row of the Cholesky decomposition of Ω_1 . By using the Lamperti transform $m(t) = \sqrt{v(t)}$, after an application of Itô's formula we

obtain

$$dm(t) = (\nu m(t)^{-1} - \rho m(t)) dt + \sum_{i=1}^4 \gamma_i dZ^{(i)}(t), \quad (23)$$

where $\nu = \frac{1}{8}(4\hat{\kappa}_4\hat{\mu}_4 - \sum_{i=1}^4 (2\gamma_i)^2)$, $\rho = \hat{\kappa}_4/2$ and $\gamma_i = \frac{1}{2}\Omega_1^{1/2(i)}(4)$, $\Omega_1^{1/2(i)}(4)$ denoting the i th entry of $\Omega_1^{1/2}(4)$, $i = 1, 2, 3, 4$. Following Kelly and Lord (2023), let us consider the mesh $\{t_0, t_1, \dots, t_M\}$ consisting of $M + 1$ equally spaced points with step size h on the interval $[0, T]$, where $t_0 = 0$ and $t_M = T$. Thus, $t_n = nh$ for $n = 0, \dots, M$ with $h = T/M$. We approximate (23) using the Lie-Trotter splitting method (see Chapter 2 of Hairer *et al.* 2006):

$$m(t+h) = e^{-\rho h} \left(\sqrt{m(t)^2 + 2\nu h} + \sum_{i=1}^4 \gamma_i \Delta Z^{(i)}(t+h) \right),$$

where $\Delta Z^{(i)}(t+h) = Z^{(i)}(t+h) - Z^{(i)}(t) \sim N(0, h)$. Then, since $\nu(t+h) = m^2(t+h)$, we compute

$$\nu(t+h) = e^{-2\rho h} \left(\sqrt{\nu(t) + 2\nu h} + \sum_{i=1}^4 \gamma_i \Delta Z^{(i)}(t+h) \right)^2,$$

or, equivalently,

$$\begin{aligned} \nu(t+h) = & e^{-\hat{\kappa}_4 h} \left(\nu(t) + 2\nu h + 2\sqrt{\nu(t) + 2\nu h} \right. \\ & \times \left(\sum_{i=1}^4 \gamma_i \Delta Z^{(i)}(t+h) \right) \\ & \left. + \left(\sum_{i=1}^4 \gamma_i \Delta Z^{(i)}(t+h) \right)^2 \right), \end{aligned}$$

which is non-negative for any $\nu > 0$. We finally proceed to derive the first (conditional) moment as in equation (16) of Kelly and Lord (2023):

$$\begin{aligned} \nu(t+h|t) = & e^{-\hat{\kappa}_4 h} \left(\nu(t) + h \left(2\nu + \sum_{i=1}^4 \gamma_i^2 \right) \right) \\ = & e^{-\hat{\kappa}_4 h} (\nu(t) + h\hat{\kappa}_4\hat{\mu}_4), \end{aligned} \quad (24)$$

which we use as a state transition equation for the variance process of the *SRV-4f* model. We note that the Kalman filtering procedure for the benchmark models remains unchanged.

5. Empirical results

Having introduced the models, the data and the estimation methodology, we proceed to describe the calibration results for market data, both in-sample and out-of-sample.

5.1. Data description

We use daily observations of crude oil futures prices sourced from Thomson Reuters Eikon Datastream. In particular, we consider light crude oil futures prices quoted on the New York Mercantile Exchange (NYMEX). Specifically, the underlying asset of the futures prices is the West Texas Intermediate (WTI) crude oil spot price, quoted in USD per barrel (USD/BBL)[†].

Each futures contract is defined by its delivery month, and its trading terminates three business days before the first day of its delivery month. The maturity date for each futures contract is set to the midweek of its delivery month. Following Schwartz (1997), we select 11 time series of futures prices, organized by their time to maturity: $F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}$, and F_{11} . Here, F_n represents the contract that is the n th month closest to maturity. The sampling period spans 24 years, from January 2000 to April 2024, encompassing a total of 6,333 business days. In figure 1, we display the time series of crude oil futures' daily prices for the maturities we considered. We note that on April 20, 2020, both the crude oil spot price and the price of the one-month maturity futures contract F_1 were quoted at negative values. This unprecedented event occurred as the COVID-19 pandemic caused a sharp decline in global petroleum demand, while U.S. crude oil inventories surged. As a result, we opted to exclude F_1 from the analysis.

Following Schwartz (1997), the bond yield data we use consist of U.S. Treasury Par Yield Curve Rates gathered from the U.S. Department of the Treasury dataset. We consider the bond yields for two maturities, 3 months and 6 months, denoted as R_3 and R_6 , respectively. In figure 2, we display the time series for these bond yields.

5.2. In-sample analysis

Table 1 presents the results of the estimation of the *SRV-4f* model. First, we perform an estimation using only futures data, applying equations (13), (15), and (24) with even maturities, specifically F_2, F_4, F_6, F_8 , and F_{10} (so that futures with odd maturities are left for the out-of-sample analysis). As already mentioned, we include bond yield data in the estimation, using equations (15), (21), and (24). In this joint estimation, bond yields R_3 and R_6 are incorporated alongside the futures prices. In both estimations, the average return on the spot commodity, $\hat{\mu}_1$, and the average convenience yield, $\hat{\mu}_2$, are positive and significant at standard levels, and the speed-of-adjustment coefficients are significant.

As expected, the correlation between the first two state variables - the spot price and convenience yield - is positive. The correlation between the spot price and volatility is also positive and significant. Thus, several correlations among the state variables are significant, contrary to the assumptions made in Yan (2002). Only the correlation between the interest rate and volatility is not significant at any conventional level. Furthermore, it is noteworthy that the significant correlation between the spot price and interest rate changes sign when bond yields are included in the estimation. This suggests that the value of

[†] BBL is an abbreviation for oilfield barrel.

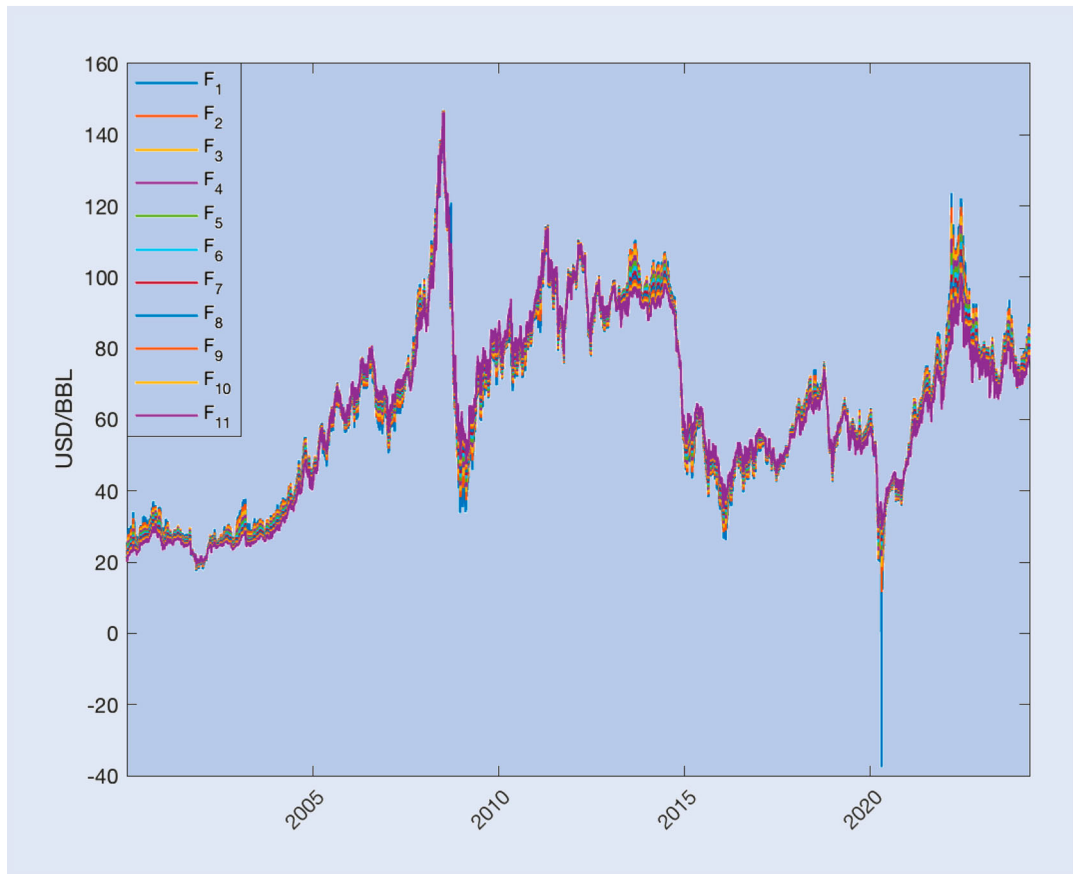


Figure 1. Crude oil futures daily prices (dollars per barrel) for several maturities from January 3, 2000 to April 14, 2024.

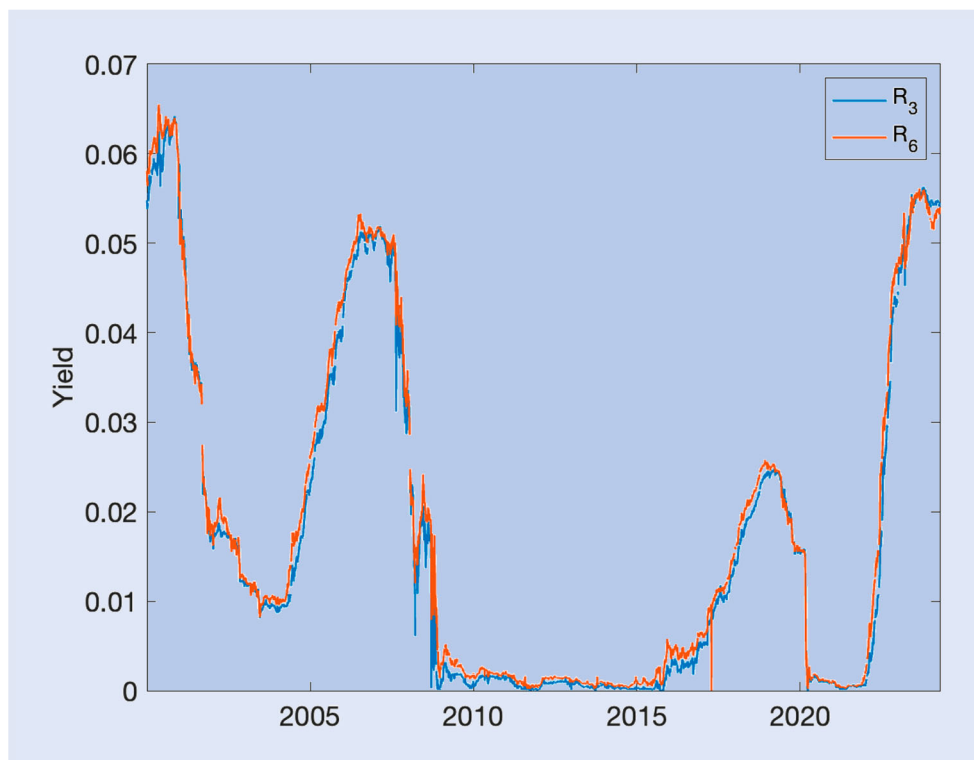


Figure 2. Bond yield daily quotes for two maturities from January 3, 2000 to April 14, 2024.

Table 1. Model estimation results for crude-oil futures.

Parameter	θ_F : Only futures	θ_{FB} : Futures and bonds
$\hat{\mu}_1$	1.670 (0.543)	0.622 (0.160)
μ_2	4.962 (0.631)	− 0.151 (0.002)
$\hat{\mu}_2$	1.989 (0.601)	0.244 (0.107)
μ_3	2.027 (0.561)	0.317 (4.000e − 2)
$\hat{\mu}_3$	2.051 (0.527)	− 0.053 (0.013)
μ_4	− 2.763 (0.221)	1.577 (8.000e − 3)
$\hat{\mu}_4$	0.227 (0.015)	0.095 (0.004)
κ_2	0.222 (0.022)	1.075 (0.011)
$\hat{\kappa}_2$	− 0.723 (0.129)	1.516 (0.267)
κ_3	2.712 (0.129)	0.030 (4.000e − 4)
$\hat{\kappa}_3$	4.711 (0.398)	− 0.022 (0.006)
κ_4	1.557 (0.085)	4.945 (0.027)
$\hat{\kappa}_4$	2.075 (0.329)	1.097 (0.221)
s_{12}	− 0.055 (0.008)	0.045 (0.002)
s_{22}	0.019 (0.004)	0.249 (0.038)
s_{13}	− 0.171 (0.018)	0.001 (1.000e − 4)
s_{23}	0.102 (0.018)	0.001 (1.000e − 4)
s_{33}	0.772 (0.053)	0.006 (3.000e − 4)
ρ_{12}	0.998 (0.056)	0.572 (0.009)
ρ_{13}	0.444 (0.113)	− 0.445 (0.020)
ρ_{14}	0.725 (0.082)	0.385 (0.017)
ρ_{23}	0.827 (0.124)	− 0.933 (0.012)
ρ_{24}	0.583 (0.081)	0.330 (0.043)
ρ_{34}	− 0.001 (2.000e − 4)	0.568 (0.317)
σ_{22}	0.375 (0.016)	0.309 (0.002)
σ_{33}	0.595 (0.146)	0.015 (0.001)
σ_{44}	0.194 (0.015)	0.155 (0.007)
σ_{ε_2}	0.011 (0.001)	0.017 (0.001)
σ_{ε_4}	1.582e − 5 (6.453e − 6)	4.000e − 3 (7.145e − 5)
σ_{ε_6}	1.000e − 3 (1.498e − 5)	6.881e − 5 (4.807e − 6)
σ_{ε_8}	6.948e − 5 (1.227e − 5)	2.000e − 4 (2.530e − 6)
$\sigma_{\varepsilon_{10}}$	3.539e − 5 (1.764e − 5)	2.000e − 3 (6.936e − 5)
σ_{ψ_3}		7.535e − 6 (1.964e − 6)
σ_{ψ_6}		2.000e − 3 (3.467e − 5)

The estimation data include crude oil futures F_2, F_4, F_6, F_8 , and F_{10} . The standard errors, reported in parentheses, are computed by inverting the negative Hessian matrix evaluated at the optimum parameter values.

Table 2. In-sample model estimation results for crude-oil futures F_2, F_4, F_6, F_8 , and F_{10} .

Model	Log.Lik.	AIC	BIC
<i>SCH-1f</i>	66,582	− 133,150	− 133,070
<i>SCH-2f</i>	85,473	− 170,920	− 170,810
<i>SCH-3f</i>	92,550	− 185,070	− 184,930
<i>HU-3f</i>	100,550	− 201,030	− 200,790
<i>YAN-4f</i>	82,786	− 165,530	− 165,350
<i>SS-4f</i>	109,670	− 219,408	− 219,612
<i>SRV-4f</i>	135,791	− 271,512	− 271,245

a futures contract is sensitive to the interest rate used in its calculation.

In table 2, we report in-sample results for the different models we considered. The information criteria suggest that the four-factor models, with the exception of the model in Yan (2002), provide a better fit to the futures term structure than one-, two-, and three-factor models. Moreover, the proposed *SRV-4f* model, when estimated solely on futures data, outperforms the other specifications.

According to equations (3)–(4), we adopt a rich parametrization for the risk premia, allowing them to depend on the full state vector $X(t)$. For comparison, we also estimated a reduced version of the *SRV-4f* model in which the dependence of the risk-premia vector $\Lambda(t, X(t))$ on the state vector $X(t)$ was restricted. Specifically, in equation (3) we imposed $\hat{B} = B$. This reduced model was calibrated using only futures data and yielded a log-likelihood of 126,280, which is substantially lower than that obtained under the full risk-premia specification (see table 2). This result indicates that the richer parametrization in equations (3)–(4) provides a marked improvement in in-sample fit.

Figure 3 presents the filtered state variables of the *SRV-4f* model corresponding to the estimation in table 1. In this figure, we also include the spot price, which is observed in the market, as well as the other state variables computed using different models or formulas. In particular, the convenience yield has been estimated using the (closed-form) formula in Javaheri *et al.* (2003), while the instantaneous interest rate has been estimated using the Vasicek model. Finally, as in Schöne and Spinler (2017), the volatility of the logarithm of the spot price has been filtered using the (annualized) volatility estimate of the GARCH(1,1) model on log-spot prices, excluding the negative price value from April 20, 2020 (see section 5.1).

From the first plot in figure 3, we observe that the first state variable closely follows the observed crude oil spot price. In the second and third plots, the filtered estimates of the convenience yield and interest rate are not as accurate when compared to the estimates from the models or functions we considered. Finally, the volatility estimate is centered around the mean of the GARCH estimate, but with lower variation.

Figure 4 mirrors figure 3 but it represents results from the joint estimation procedure, using crude oil futures F_2, F_4, F_6, F_8 , and F_{10} in conjunction with bond yields R_3 and R_6 . In this case, we can see that the filtered estimate of the interest rate is closer to the Vasicek model estimate. The convenience yield estimate is also more aligned with the one from Javaheri *et al.* (2003), resulting in a more accurate representation of the spot price drift term dynamics.

In figure 5, we display the prediction errors $e(t)$ for both the futures-only estimation and the joint estimation using futures and bond yields. As we may see, the errors are centered around zero and exhibit volatility clustering, indicating changing levels of uncertainty over time.

5.3. Out-of-sample analysis

As is standard in the reference literature (see Schwartz 1997, Schöne and Spinler 2017), we evaluate futures prices for maturities not included in the estimation. We consider futures prices corresponding to odd maturities, that is, F_3, F_5, F_7, F_9 , and F_{11} , and we evaluate them using the parameters from table 1. More precisely, after filtering the state variables and obtaining the coefficients in the futures pricing equation (8), we can price the futures that were excluded from the estimation, that is F_3, F_5, F_7, F_9 , and F_{11} . We then compare models using standard metrics, namely root mean square error (RMSE) and mean absolute percentage error (MAPE). For a futures contract with maturity T , the error metrics are defined

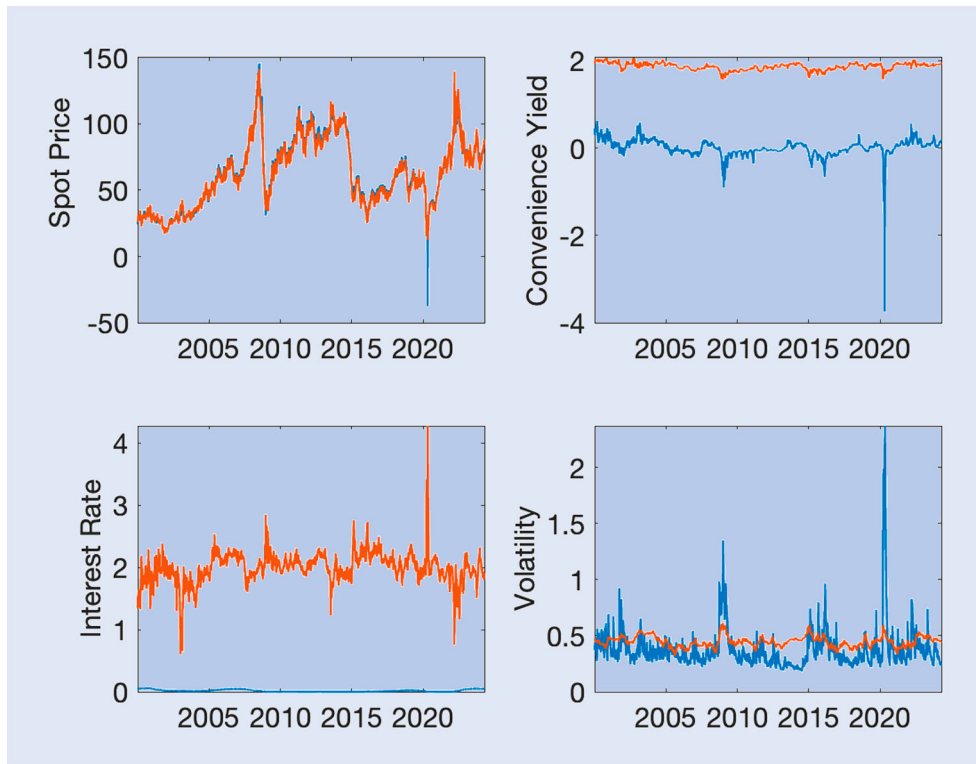


Figure 3. Filtered (red) and observable (blue) state variables (futures-only estimation).



Figure 4. Filtered (red) and observable (blue) state variables (joint futures-bond estimation).

as follows:

$$\text{RMSE}(T) = \sqrt{\frac{\sum_{t=1}^N (y(t, T) - \tilde{y}(t, T, \tilde{X}(t)))^2}{N}},$$

$$\text{MAPE}(T) = \frac{1}{N} \sum_{t=1}^N \frac{|y(t, T) - \tilde{y}(t, T, \tilde{X}(t))|}{y(t, T)},$$

where N is the length of the time series of futures prices, $y(t, T)$ is the observed log futures price at time t with maturity

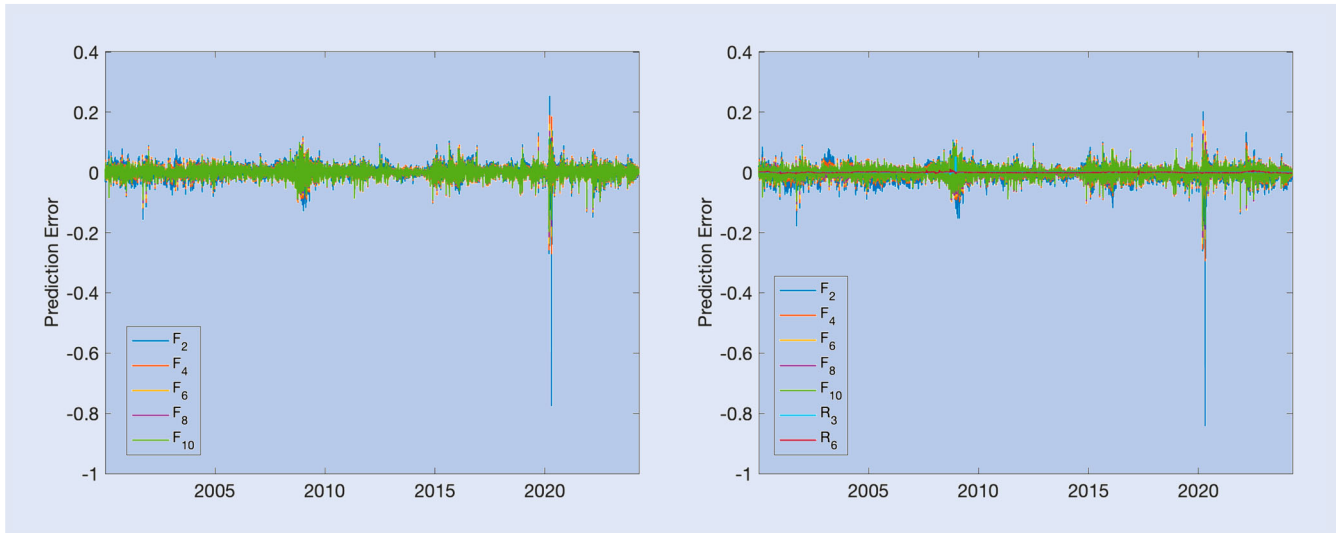


Figure 5. Prediction errors using futures (left panel) and using both futures and bond yields (right panel). The left panel shows errors from the estimation based solely on futures, while the right panel reflects errors from the joint estimation.

Table 3. Out-of-sample RMSE and MAPE pricing errors (in percentage) for valuing futures contracts for the considered models, along with the out-of-sample predictive log-likelihood in equation (18), computed using the $F_3, F_5, F_7, F_9, F_{11}$ futures. Estimation is performed using futures $F_2, F_4, F_6, F_8, F_{10}$ and bond yields R_3 and R_6 . $SRV-4f$ (θ_F) and $SRV-4f$ (θ_{FB}) denote the estimations using only futures and using both futures and bonds, respectively.

Futures	<i>SCH-1f</i>	<i>SCH-2f</i>	<i>SCH-3f</i>	<i>HU-3f</i>	<i>YAN-4f</i>	<i>SS-4f</i>	<i>SRV-4f</i> (θ_F)	<i>SRV-4f</i> (θ_{FB})
RMSE(3)	4.588	2.095	1.332	1.717	2.241	1.577	1.568	1.436
RMSE(5)	2.964	1.097	1.256	1.477	1.120	1.195	1.118	1.064
RMSE(7)	2.667	0.891	1.041	1.118	0.994	0.956	0.904	0.906
RMSE(9)	3.339	0.923	0.788	0.768	0.846	0.789	0.773	0.773
RMSE(11)	4.324	1.163	0.913	0.721	0.715	0.708	0.686	0.745
Mean	3.576	1.234	1.066	1.160	1.183	1.045	1.010	0.985
MAPE(3)	0.873	0.440	0.254	0.287	0.522	0.314	0.319	0.303
MAPE(5)	0.555	0.224	0.235	0.272	0.231	0.240	0.227	0.227
MAPE(7)	0.489	0.189	0.204	0.224	0.204	0.196	0.188	0.190
MAPE(9)	0.655	0.183	0.166	0.164	0.176	0.165	0.165	0.165
MAPE(11)	0.889	0.233	0.192	0.149	0.154	0.148	0.148	0.159
Mean	0.692	0.254	0.210	0.219	0.257	0.213	0.209	0.209
Predictive Log.Lik.	68,631	86,761	93,987	107,420	83,555	112,160	138,862	126,680

T and $\tilde{y}(t, T, \tilde{X}(t)) = \ln F(t, T - t, \tilde{X}(t))$ is the corresponding predicted value, with $\tilde{X}(t)$ representing the filtered state variable. For the *SRV-4f* model, we evaluate performance for both the futures-only estimation and the joint estimation involving futures and bonds. In the case of joint estimation, when calculating the predictive likelihood, we used only the futures data, while the model parameters were estimated using both futures and bond data. The results are reported in table 3, where we also include the predictive log-likelihood, which is the log-likelihood function evaluated on out-of-sample data using in-sample parameters (for more details, see González-Rivera *et al.* 2004).

As we may see, the *SRV-4f* model demonstrates outstanding performance in both the futures-only estimation and the joint estimation with futures and bonds. Specifically, for the futures-only estimation, the RMSE and MAPE futures pricing errors are lower for almost all maturities compared to other models, and the predictive likelihood value is higher. When estimated jointly with bond data, the *SRV-4f* model achieves the lowest average errors for shorter maturities, likely due to the inclusion of short-term bonds in the estimation process.

To assess whether the performance of our proposed model is influenced by short-maturity futures, we re-estimate the *SRV-4f* model using contracts F_4, F_6, F_8, F_{10} , and F_{12} , that is, replacing the two-month maturity future with the twelve-month contract. Again, the out-of-sample performance is evaluated using the odd-maturity futures F_3, F_5, F_7, F_9 , and F_{11} . The results are reported in table 4.

As shown in table 4, the *SRV-4f* model continues to exhibit strong out-of-sample performance even when short-maturity futures are excluded from the estimation. Overall, the results obtained using F_4, F_6, F_8, F_{10} , and F_{12} in the estimation process are broadly consistent with those reported when the model is estimated with F_2, F_4, F_6, F_8 , and F_{10} . However, unlike the results in table 3, the joint estimation using both futures and bonds no longer provides a clear overall advantage: the futures-only estimation achieves slightly lower average RMSE and MAPE values (0.932 and 0.188) compared to the joint case (1.024 and 0.203).

Despite this, the joint estimation delivers better performance for the longer maturities F_9 and F_{11} , indicating that the inclusion of bond data contributes to a more accurate

Table 4. Out-of-sample performance of the *SRV-4f* model when estimation is performed using futures $F_4, F_6, F_8, F_{10}, F_{12}$ and bond yields R_3 and R_6 . *SRV-4f* (θ_F) and *SRV-4f* (θ_{FB}) denote the estimations using only futures and using both futures and bonds, respectively.

Futures	<i>SRV-4f</i> (θ_F)	<i>SRV-4f</i> (θ_{FB})
RMSE(3)	1.432	1.822
RMSE(5)	1.004	1.100
RMSE(7)	0.831	0.831
RMSE(9)	0.737	0.726
RMSE(11)	0.653	0.639
Mean	0.932	1.024
MAPE(3)	0.260	0.331
MAPE(5)	0.208	0.220
MAPE(7)	0.176	0.176
MAPE(9)	0.157	0.155
MAPE(11)	0.141	0.135
Mean	0.188	0.203
Predictive Log.Lik.	141,080	129,050

characterization of the long-term factors driving the term structure. This improvement at the far end of the curve likely reflects the information content of bond yields regarding long-run expectations and risk premia, which complement the signals from long-dated futures. Interestingly, in the previous estimation setup (table 3), where shorter-maturity futures such as F_2 were included, the use of bond data mainly improved the pricing accuracy for short maturities, notably F_3 . In contrast, when estimation relies on longer-maturity contracts, the inclusion of bonds becomes more relevant for capturing the behavior of longer-term futures, such as F_9 and F_{11} . So, the joint estimation improves the forecasting of short-term futures when shorter-maturity contracts are used for calibration and enhances the forecasting of long-term futures when longer-maturity contracts are used. Overall, the results obtained confirm the robustness of the *SRV-4f* model across different estimation settings.

5.4. A hedging experiment

The proposed *SRV-4f* model can also be employed for hedging purposes. To illustrate this, we perform a simple hedging experiment in which a long position in the crude oil spot price is hedged using a corresponding futures contract. For simplicity, we consider only the futures contract with the shortest maturity (three months) and compute the minimum variance hedge ratio (MVHR), aiming to mitigate the short-term risk associated with fluctuations in the spot price of the commodity.

Our framework can be extended to design more sophisticated hedging strategies, for example by considering futures with multiple maturities, bond instruments, or second-order hedging techniques accounting for interest rate and volatility risks (see Schwartz 1997, Mellios *et al.* 2016, Cheng *et al.* 2019). A comprehensive analysis of such extensions within the *SRV-4f* framework is left for future research.

Consider a long position in the commodity hedged by taking a short position in h futures contracts on the same

commodity. Since we aim to hedge the spot price, we select the futures contract with a three-month maturity. The value of the hedged portfolio at time t is then

$$\Pi(t) = S(t) - h(t)F(t, T - t, X(t)).$$

For a small time variation (keeping $h(t)$ constant), the portfolio change is approximated by

$$\Delta \Pi = \Delta S - h \Delta F,$$

where, for the moment, dependence on t and X is omitted for brevity. According to equation (8), $\Delta \ln F \approx \beta^\top \Delta X$, and, since $X_1 = \ln(S)$ and $dX_1 = dS/S$, we have:

$$\Delta F \approx F(\beta^\top \Delta X), \quad \Delta S \approx S \Delta X_1.$$

Consequently,

$$\text{Cov}(\Delta S, \Delta F) = SF \text{Cov}(\Delta X_1, \beta^\top \Delta X)$$

$$= SF \sum_{i=1}^4 \beta_i \text{Cov}(\Delta X_1, \Delta X_i), \quad (25)$$

$$\text{Var}(\Delta F) = F^2 \text{Var}(\beta^\top \Delta X) = F^2 (\beta^\top \Sigma_X \beta), \quad (26)$$

where $\Sigma_X = \text{Cov}(\Delta X, \Delta X)$ denotes the covariance matrix of the four state variables.

Minimizing the variance of $\Delta \Pi$ yields the MVHR. Accordingly, for each trading day t , the optimal hedge ratio is computed empirically as

$$\hat{h}(t) = \frac{\text{Cov}(\Delta S, \Delta F)}{\text{Var}(\Delta F)} = \frac{S^{\text{mkt}}(t)}{F_3^{\text{mkt}}(t)} \frac{\beta^\top \Sigma_{X,1}}{\beta^\top \Sigma_X \beta}, \quad (27)$$

where $S^{\text{mkt}}(t)$ and $F_3^{\text{mkt}}(t)$ denote the observed market spot and futures prices with three-month maturity at day t , and $\Sigma_{X,1}$ is the first column of Σ_X , corresponding to the covariances between the spot-price factor ΔX_1 and the remaining state variables. The covariance matrix Σ_X is updated daily and estimated using a rolling window of 252 trading days (from $t - 251$ to t) based on the historical factor increments. The model parameters are kept fixed at the values reported in table 1.

We compare the Profit/Loss (P&L) of an unhedged position:

$$\text{PL}^{\text{unhedged}}(t) = S^{\text{mkt}}(t) - S^{\text{mkt}}(t - 1),$$

with that of a hedged portfolio:

$$\begin{aligned} \text{PL}^{\text{hedged}}(t) &= S^{\text{mkt}}(t) - S^{\text{mkt}}(t - 1) \\ &\quad - \hat{h}(t - 1)(F_3^{\text{mkt}}(t) - F_3^{\text{mkt}}(t - 1)). \end{aligned}$$

The hedge effectiveness is measured as the percentage reduction in P&L variance:

$$R^{\text{MVHR}} = 1 - \frac{\text{Var}(\text{PL}^{\text{hedged}})}{\text{Var}(\text{PL}^{\text{unhedged}})}.$$

The empirical value obtained is $R^{\text{MVHR}} = 0.71$, indicating that the simple delta-hedging strategy removes a substantial portion of the spot price risk: the variance of the hedged portfolio is approximately 30% of that of the unhedged one.

6. Conclusions

This paper introduces a new four-factor model (*SRV-4f*) for commodity pricing that jointly captures the stochastic dynamics of the spot price, convenience yield, stochastic volatility, and stochastic interest rates. While Yan (2002) also considers this full set of risk drivers, our framework extends this line of research by allowing for a richer correlation structure driven by the volatility factor, thereby generating flexible co-movements among all state variables.

Economically, the model reflects the interaction between the macroeconomic environment, proxied by the short-term interest rate, and commodity-specific fundamentals, such as the convenience yield and the volatility of the underlying asset. By allowing the stochastic volatility factor to enter the diffusion terms of all state variables, the model captures the empirical reality that volatility shocks in commodity markets are typically correlated with changes in macroeconomic uncertainty, inventory conditions, and information flows. Therefore, the proposed specification enriches the interplay between physical and financial risk channels while preserving analytical tractability and closed-form expressions for pricing futures and bonds.

We also develop a unified Kalman filter estimation procedure that jointly employs futures and bond prices. This joint estimation improves the identification of the interest-rate dynamics and the treatment of risk premia across markets, while preserving a coherent state-space representation suitable for empirical implementation.

Based on empirical evidence from NYMEX crude oil futures and U.S. Treasury yields, the *SRV-4f* model consistently outperforms established benchmarks such as Schwartz (1997), Yan (2002), HUGHEN (2010), and Schöne and Spinler (2017), in both in-sample and out-of-sample evaluations. Moreover, when the estimation relies on shorter-maturity contracts ($F_2, F_4, F_6, F_8, F_{10}$), including bond yields improves the pricing accuracy of short-term futures, particularly at the near end of the curve. Conversely, when the estimation sample is shifted toward longer-maturity contracts ($F_4, F_6, F_8, F_{10}, F_{12}$), the inclusion of bond data enhances the characterization of long-term dynamics, leading to better performance at the far end of the curve. Overall, the joint estimation confirms the robustness and flexibility of the *SRV-4f* model across different estimation settings. Furthermore, a simple hedging experiment further illustrates the model's practical relevance: the model-based minimum-variance hedge ratio reduces portfolio variance by more than 70%.

For practitioners, the model furnishes a coherent and economically grounded framework for pricing, forecasting, and hedging in environments where financial and physical commodity markets are closely integrated. More broadly, *SRV-4f* constitutes a theoretically well-founded, empirically validated, and economically interpretable structure for the joint modeling of spot prices, convenience-yield volatility, and interest-rate dynamics, as well as their combined implications for commodity-futures valuation. As such, it provides a robust foundation for research and practice in pricing, risk management, and policy analysis.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Code availability statement

The code for calibrating the *SRV-4f* model is available in MATLAB at the GitHub repository: https://github.com/tezzaachris/SRV_four_factors.

Data availability statement

The data used in this study were obtained from LSEG Workspace (Refinitiv) under institutional license and cannot be shared by the authors. Interested researchers can obtain the data directly from LSEG Workspace (Refinitiv), subject to a licence agreement.

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Appendix

In this Appendix, we report the tables containing the estimated parameters for all models used as benchmarks.

Table A1. Model estimation results for the SCH-If model (Schwartz 1997).

Parameter	
μ	3.468 (0.129)
κ	0.005 (0.001)
λ	2.863 (0.278)
σ	0.099 (0.002)

The estimation data include crude oil futures F_2 , F_4 , F_6 , F_8 , and F_{10} . The standard errors, reported in parentheses, are computed by inverting the negative Hessian matrix evaluated at the optimum parameter values. Following Schwartz (1997), $\alpha = \mu - \frac{\sigma^2}{2\kappa}$ and λ is the market price of risk.

Table A2. Model estimation results for the *SCH-2f* model (Schwartz 1997).

Parameter	
μ	1.012 (0.185)
σ_1	0.253 (0.006)
r	0.223 (0.023)
ρ	0.465 (0.016)
α	0.015 (0.007)
σ_2	0.039 (0.002)
κ	0.461 (0.076)
λ	− 0.268 (0.032)

The estimation data include crude oil futures F_2, F_4, F_6, F_8 , and F_{10} . The standard errors, reported in parentheses, are computed by inverting the negative Hessian matrix evaluated at the optimum parameter values. As in equation (15) of Schwartz (1997), λ denotes the market price of risk for the convenience yield.

Table A3. Model estimation results for the *SCH-3f* model (Schwartz 1997).

Parameter	
α	0.719 (8.683)
μ	2.952 (9.619)
κ	0.155 (0.045)
a	0.329 (0.134)
σ_1	0.343 (0.009)
σ_2	0.142 (0.065)
σ_3	0.203 (0.193)
ρ_1	0.155 (1.589)
ρ_2	0.284 (1.269)
ρ_3	0.500 (0.259)
λ_1	− 0.018 (0.045)
λ_2	− 0.032 (0.013)

The estimation data include crude oil futures F_2, F_4, F_6, F_8 , and F_{10} . The standard errors, reported in parentheses, are computed by inverting the negative Hessian matrix evaluated at the optimum parameter values. To transition from the risk-neutral measure to the physical measure, we substitute $\hat{\alpha} = \alpha - \lambda_1/\kappa$ in the equation for the convenience yield and $\hat{\mu} = \mu - \lambda_2/a$ in the equation for the interest rate.

Table A4. Model estimation results for the *HU-3f* model (Hughen 2010).

Parameter	
Risk-Neutral Drift Parameters (μ_i, k_{ij})	
μ_2	0.047 (0.209)
μ_3	0.863 (0.211)
k_{21}	0.604 (0.189)
k_{22}	-1.254 (0.209)
k_{23}	-0.878 (0.061)
k_{33}	0.478 (0.111)
Physical Drift Parameters (a_i, b_{ij})	
a_1	0.606 (16.360)
a_2	0.705 (10.593)
a_3	0.194 (1.790)
b_{11}	0.185 (0.340)
b_{12}	-0.600 (5.248)
b_{13}	-0.636 (7.076)
b_{21}	-0.470 (0.480)
b_{22}	-1.236 (7.125)
b_{23}	-0.257 (0.813)
b_{33}	-1.109 (1.479)
Volatility and Correlation Parameters	
s_{12}	0.187 (0.688)
s_{22}	0.427 (0.051)
ϑ	-0.020 (0.078)
ρ_{12}	-0.648 (0.201)
ρ_{13}	-0.400 (0.337)
ρ_{23}	0.161 (0.116)
σ_{22}	0.009 (0.004)
σ_{33}	0.161 (0.033)

The estimation data include crude oil futures F_2, F_4, F_6, F_8 , and F_{10} . The standard errors, reported in parentheses, are computed by inverting the negative Hessian matrix evaluated at the optimum parameter values. Note: $\sigma_{12} = \rho_{12}\sigma_{22}$, $\sigma_{13} = \rho_{13}\sigma_{33}$ and $\sigma_{23} = \rho_{23}\sigma_{22}\sigma_{33}$.

Table A5. Model estimation results for the *YAN-4f* model (Yan 2002).

Parameter	
ν	0.590 (0.021)
μ_J	0.132 (0.001)
σ_J	0.803 (0.010)
θ	0.100 (0.001)
σ_x	0.834 (0.006)
μ_r	0.355 (0.001)
k_r	0.760 (7.44×10^{-6})
σ_r	0.000 (1.38×10^{-9})
λ_r	0.001 (0.002)
μ_δ	0.126 (0.013)
k_δ	0.598 (1.35×10^{-4})
σ_δ	0.448 (0.002)
λ_δ	0.004 (0.009)
μ_v	0.320 (0.018)
σ_v	0.075 (0.006)
λ_v	0.000 (0.001)
k_v	0.082 (0.014)
$\rho_{x\delta}$	0.146 (0.012)
ρ_{xv}	0.300 (0.057)

The estimation data include crude oil futures F_2, F_4, F_6, F_8 , and F_{10} . The standard errors, reported in parentheses, are computed by inverting the negative Hessian matrix evaluated at the optimum parameter values. To transition from the risk-neutral measure to the physical measure, we substitute $\hat{\mu}_\delta = \mu_\delta - \lambda_\delta$ in the equation for the convenience yield, $\hat{\mu}_r = \mu_r - \lambda_r$ in the equation for the interest rate, and $\hat{\mu}_v = \mu_v - \lambda_v$ in the equation for the variance.

Table A6. Model estimation results for the *SS-4f* model (Schöne and Spinler 2017).

Parameter	
μ_2	0.292 (0.003)
μ_3	1.288 (0.009)
μ_4	1.529 (0.005)
$\hat{\mu}_1$	0.281 (0.006)
$\hat{\mu}_2$	0.355 (0.002)
$\hat{\mu}_3$	1.225 (0.004)
$\hat{\mu}_4$	1.568 (0.006)
κ_{11}	0.240 (0.000)
κ_{33}	1.222 (0.000)
κ_{14}	0.174 (0.001)
κ_{44}	0.007 (0.000)
$\hat{\kappa}_{11}$	0.307 (0.000)
$\hat{\kappa}_{33}$	1.773 (0.000)
$\hat{\kappa}_{44}$	0.286 (0.001)
s_{12}	0.056 (0.000)
s_{13}	0.009 (0.000)
s_{22}	0.021 (0.000)
s_{23}	0.013 (0.000)
s_{33}	0.324 (0.000)
ϑ	-0.018 (0.001)
ρ_{12}	0.744 (0.001)
ρ_{13}	-0.052 (0.000)
ρ_{14}	0.101 (0.000)
ρ_{23}	0.119 (0.000)
ρ_{24}	0.040 (0.000)
ρ_{34}	0.116 (0.000)
σ_{22}	0.029 (0.000)
σ_{33}	0.013 (0.000)
σ_{44}	0.005 (0.000)

The estimation data include crude oil futures F_2 , F_4 , F_6 , F_8 , and F_{10} . The standard errors, reported in parentheses, are computed by inverting the negative Hessian matrix evaluated at the optimum parameter values. Values are rounded to the third decimal place. Note: $\sigma_{12} = \rho_{12}\sigma_{22}$, $\sigma_{13} = \rho_{13}\sigma_{33}$, $\sigma_{14} = \rho_{14}\sigma_{44}$, $\sigma_{23} = \rho_{23}\sigma_{22}\sigma_{33}$, $\sigma_{24} = \rho_{24}\sigma_{22}\sigma_{44}$, and $\sigma_{34} = \rho_{34}\sigma_{33}\sigma_{44}$. The parameters with a hat ($\hat{\cdot}$) are used to compute risk premia, as in equations (28) and (30) of (Schöne and Spinler 2017).